act like a senior Front-end Engineer. I want to make my portfolio. I want this section Home, About, Skills, Professional Experience, Blog, Contact and Faq. I'm Towhid Ahmed. I'm a Generative AI Engineer. I give you 1 picture. Picture is attach with home page.

Home section write: Generative AI Engineer | Agentic AI & LLMOps | Advanced RAG Systems | LLM Applications | Open to Remote Opportunities. Add 1 beautiful button Generative AI Engineer CV. When user click resume button open new window this link: https://genai-resume-seven.vercel.app

About section: you have to write:
Generative AI Engineer | Agentic Systems | LLMOps | Advanced RAG
I am Towhid Ahmed, a Generative AI Engineer who independently architects and deploys production-grade intelligent systems. I specialize in building end-to-end LLM applications with a focus on multi-agent orchestration, advanced Retrieval-Augmented Generation (RAG), and LLMOps—from concept to deployment. My approach centers on solving real-world problems through scalable AI: designing sophisticated agent workflows using LangGraph, implementing retrieval systems with vector databases and cross-encoder re-ranking, and ensuring production reliability through observability, testing, and guardrails. I'm actively seeking opportunities to contribute to innovative AI teams where I can help build and ship scalable, intelligent products that make a tangible impact. I thrive in collaborative, fast-paced environments and bring strong systems thinking, 400+ DSA problems solved, and a proven track record of turning ambitious ideas into deployed systems.

TECHNICAL SKILLS: you have to write :

Generative AI & Agentic Systems
LangChain · LangGraph · Multi-Agent Orchestration · Tool Calling · Structured Outputs · Prompt Engineering · Advanced RAG (Agentic RAG, Multimodal RAG, Corrective RAG, Graph RAG) · RAG Pipeline Design · Retrieval Strategies · Cross-Encoder Re-ranking · Vector Databases (FAISS, ChromaDB, Weaviate, Pinecone)

LLMOps & Production Deployment
LangSmith · LLM Guardrails · FastAPI · REST APIs · Async Architecture · Docker · Streamlit · Observability & Monitoring · CI/CD (GitHub Actions) · Render

Large Language Models & AI Frameworks
OpenAI · Groq · LLaMA · Mistral · Transformers · BERT · Sentence Transformers · Hugging Face · Whisper

Machine Learning & Deep Learning
Scikit-learn · TensorFlow · CNNs · RNNs/LSTMs · Supervised & Unsupervised Learning · MLflow · Feature Engineering

Backend & Data
Python · Java · PostgreSQL · SQL · Pandas · NumPy · SQLite · APScheduler

Development & Version Control
Git · GitHub · Object-Oriented Programming (OOP) · Design Patterns · Testing (Pytest) · System Design

Core Fundamentals
Data Structures & Algorithms (400+ LeetCode problems solved) · LLM System Design · Production-Grade Code Architecture

PROFESSIONAL EXPERIENCE : In this section you write experience project based.

I give here my project details with github and demo link. You have to amazing this section with your best. In project you have to show only projects title. Then Add a something like button in every project. When i click the button my project details open.

Multi-Agent AI System — Python, LangGraph, FastAPI, Pinecone, LangSmith, Docker, Render

● Architected a 4-agent LangGraph workflow (Planner, Retrieval, Search, Synthesizer) with production FastAPI backend (/chat, /upload, /health, /metrics) supporting async-safe execution for reliable cloud deployment.
● Built end-to-end RAG pipeline: document ingestion (PDF/TXT/Markdown) → chunking → Pinecone vector retrieval → citation synthesis, integrated with Tavily + DuckDuckGo for grounded, up-to-date responses.
● Added production observability: LangSmith tracing, structured logging, and in-process latency/error metrics for debugging and workflow evaluation.
● Deployed via Docker + Render Blueprint with health checks and optimizations suitable for free-tier, remote-accessible hosting.

https://github.com/TowhidAhmedd/multi-agent-AI-Orchestrator/tree/main
https://multi-agent-ai-system-frontend-with.onrender.com

Multimodal RAG — Python, FastAPI, Streamlit, Pinecone, Groq, Llama 3.3, Faster-Whisper, RAG, Vector Search, LangSmith, Docker, REST API, Async Programming, Caching
● Architected production-grade multimodal RAG system (PDF, audio, video) with Faster-Whisper transcription and semantic chunking; implemented two-stage retrieval (Pinecone + Cross-Encoder re-ranking), reducing latency by 60% and improving relevance scores by 25%.
● Built async FastAPI backend with Groq LLM integration (3 model options), TTL-based query caching, and structured error handling; achieved <2s P95 latency on cached queries with 100% test pass rate (76 unit tests).
● Deployed full-stack application on Render with Docker containerization (FastAPI + Streamlit), automated CI/CD via GitHub, and LangSmith observability for production monitoring.

https://github.com/TowhidAhmedd/Multimodel-Rag-With-LLMOps/tree/main

Clinical RAG Assistant — Python, FastAPI, LangGraph, RAG, Pinecone, LLM Guardrails, Tavily, Streamlit, LangSmith, Docker, JWT
● Architected production-grade medical education AI: 4-agent LangGraph workflow (Router → Safety → Retrieval → Answer) with automatic fallback between WEB_ONLY, DOC_ONLY, and HYBRID search modes.
● Implemented RAG pipeline: PDF/DOCX/TXT ingestion → BAAI/bge embeddings → Pinecone retrieval → FlashRank re-ranking → page-level citations with domain whitelist (NIH, CDC, WHO, Mayo Clinic).
● **Engineered industry-leading 4-layer medical safety system: (1) regex input filter, (2) LLM safety agent, (3) retrieval threshold guard, (4) output regex filter—blocking diagnosis, dosage, and injection attacks.**
● Built secure FastAPI backend with JWT auth, API key validation, SlowAPI rate limiting, Pydantic v2 validation, and multipart file enforcement.
● Delivered Streamlit frontend with real-time health checks, document management UI, confidence-scored citations, and graceful error handling.

https://github.com/TowhidAhmedd/Clinical-AI/tree/main

Advanced RAG System — Python, LangChain, LLM APIs, Vector Database, Streamlit, RAG
● Designed an **LLM-based retrieval system and AI research assistant for semantic document search and QA over ML research papers**
● Built an **end-to-end retrieval-augmented generation pipeline covering PDF ingestion, intelligent chunking, embedding generation, Weaviate vector search, and LLM-based response generation**

● Designed a **modular architecture separating ingestion, retrieval, LLM orchestration, and UI layers** with **context-grounded responses** to improve **answer reliability and reduce hallucinations**.

https://github.com/TowhidAhmedd/Advanced_Rag_ML_Research_Assistant/tree/main

Autonomous Deep Agent Platform - Python FastAPI LangGraph LangChain Neo4j Pinecone Groq Streamlit Docker

● Architected an 8-node LangGraph agent graph with conditional routing, retry loops, and human-in-the-loop checkpoints that autonomously plans, researches, retrieves, executes, and self-verifies answers to complex multi-step goals.
● Built world-class hybrid RAG pipeline combining Pinecone semantic vector search with Neo4j graph traversal—extracting entities/relationships via LLM, storing as graph nodes/edges, then fusing both retrieval paths at query time for relationship-aware intelligence.
● Implemented Verifier Agent scoring answers on accuracy, completeness, relevance, and clarity; detects hallucinations by comparing claims against grounding context and triggers automated retry loops when quality thresholds unmet.
● Shipped production infrastructure: multi-stage Docker, Docker Compose orchestrating FastAPI + Streamlit + Neo4j, LangSmith observability, multi-provider LLM support (Groq, OpenAI, Gemini), and 93 passing tests.

https://github.com/TowhidAhmedd/agent-deepagent/tree/main

AI News Agent - LangGraph · LangChain · FastAPI · APScheduler · SQLite · Docker · Groq
● Built fully automated AI news intelligence system using 8-node LangGraph multi-agent pipeline (Discovery → Extraction → Scoring → Dedup → Summarization → Ranking → Newsletter → Email) delivering formatted HTML briefing emails daily at 6:30 AM, 100% free-tier.
● Engineered stateful async LangGraph workflow with 8 independently error-resilient agents across 18 RSS sources (OpenAI, Anthropic, DeepMind, VentureBeat, MIT Tech Review, etc.), using parallel batch scoring (5 articles/call, 3 parallel calls) to minimize latency and token costs.
● Implemented two-stage deduplication (URL normalization + Jaccard title similarity) eliminating syndicated duplicates without LLM calls—cutting downstream processing costs significantly; built provider-agnostic LLM factory (Groq/OpenRouter/Gemini) enabling zero-code switching.
● Deployed production system via Docker Compose with APScheduler cron, FastAPI dashboard (/run-now, /newsletter/latest, /metrics), LangSmith distributed tracing, SQLite persistence, and 32/32 test coverage (unit, integration, end-to-end).

https://github.com/TowhidAhmedd/agent-ai-news/blob/main/resume.txt

Human-in-the-Loop AI Email Reply Agent - LangGraph · FastAPI · Groq (LLaMA 3.3-70B) · Gmail API · ChromaDB · Streamlit · Docker

● Architected a 7-node LangGraph stateful pipeline (fetch → classify → context retrieval → draft → safety review → human approval → send) with hard interrupt/resume mechanism enforcing zero auto-sends—no email leaves without explicit human approval.
● Engineered multi-layer HITL safety across three independent enforcement points (routing edge, node-level gate, API validation) so no single failure mode can bypass human control; implemented LLM-powered safety review agent scoring drafts for hallucinations, incorrect promises, sensitive data leakage, and aggressive language.
● Built ChromaDB vector memory system retrieving semantically similar past threads, approved drafts, and company knowledge to generate context-aware, personalized replies using cosine similarity search.
● Delivered full-stack Streamlit approval dashboard with inline draft editing, one-click approve/reject, risk score display, and agent analytics; deployed via Docker Compose with APScheduler, SQLite persistence, and 80%+ test coverage.

https://github.com/TowhidAhmedd/agent-email-replay/tree/main

Blog:

Generative AI Explained: How Modern AI Creates Text, Images, and Code
Deep dive into LLMs, Transformers, Diffusion Models, and real-world AI systems.

Read Blog:
https://medium.com/@towhidahmed422/generative-ai-explained-how-modern-ai-creates-text-images-and-code-complete-guide-for-d663e893e600

Tags: Generative AI | LLM | Machine Learning

How I Approach Building Real-World AI Systems as an ML Engineer

Read Blog:
https://medium.com/@towhidahmed422/how-i-approach-building-real-world-ai-systems-as-an-ml-engineer-e99abfc9b452

Tags: Machine Learning AI Engineering MLOps System Design FastAPI

Building an End-to-End Machine Learning Project — From Data to Deployment

Read Blog:
https://medium.com/@towhid4635/building-an-end-to-end-machine-learning-project-from-data-to-deployment-5213707a4d64
Tags: Machine Learning FastAPI MLOps Python

Understanding RAG (Retrieval-Augmented Generation) — The Future of LLM Applications

Read Blog:
https://medium.com/@towhid4635/understanding-rag-retrieval-augmented-generation-the-future-of-llm-applications-85ebad32d0c8
Tags: **RAG (Retrieval Augmented Generation) Large Language Models (LLMs) Vector Databases Embeddings**

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Email, message, submit button

FAQ:
1. Question: Tell me about yourself

Answer: I’m Towhid Ahmed, a Generative AI Engineer focused on building production-grade LLM applications, Agentic AI systems, and advanced RAG solutions. I specialize in designing end-to-end AI systems using LangGraph, LangChain, vector databases, and LLMOps practices — from retrieval pipelines to deployment and monitoring.

I have built real-world AI solutions including multi-agent systems, autonomous agents, multimodal RAG applications, and intelligent AI assistants. With strong software engineering fundamentals and 400+ DSA problems solved, I combine AI expertise with scalable system design to build reliable, impactful intelligent products.

2. Question: Are you working in any timezone?
Answer: I am based in Bangladesh (GMT+6), but I am flexible and comfortable working in different time zones depending on team requirements.
3. Question: Have you worked with any team?
Answer: I have primarily worked independently on my projects, which helped me develop strong self-learning and problem-solving skills. However, I am comfortable collaborating with teams and using tools like GitHub for version control and project management.
4. Question: Are you open to remote work?
Answer: Yes, I am open to remote opportunities and comfortable working in distributed teams.
5. Question: What tools do you use for collaboration?
Answer: I use Git and GitHub for version control and collaboration. I am also familiar with communication tools like Slack, Trello, and Jira.
6. Question: How do you manage your time?
Answer: I prioritize tasks based on importance and deadlines. I break down work into smaller tasks and follow a structured approach to complete projects efficiently.
7. Question: Are you comfortable learning new technologies?
Answer: Yes, I actively learn and explore new technologies, with a strong focus on Generative AI, LLMs, agentic systems, RAG architectures, and emerging AI tools. I enjoy quickly adapting to new technologies and applying them to build practical, real-world solutions.

8. Question: Why should we hire you?
Answer: I believe you should hire me because I have hands-on experience designing and building end-to-end Generative AI systems, including LLM applications, RAG pipelines, and agent-based workflows. I combine strong problem-solving fundamentals with the ability to independently learn, build, and deploy real-world AI solutions. I am highly motivated to contribute to an AI team, solve challenging problems, and continuously improve my technical skills.

Create a floating contact button section for a portfolio website.
Requirements:
- Add two circular floating buttons fixed at the bottom of the screen.
-Rightside button: WhatsApp logo with link to my WhatsApp number.
- Right side button: LinkedIn logo with link to my LinkedIn profile.
- Both buttons should always stay visible (position: fixed).
- Add hover effect (slight zoom or scale).
- Buttons should have shadow and modern UI design.
- Make it responsive for mobile and desktop.
WhatsApp link format: https://wa.me/your-number
LinkedIn link: https://linkedin.com/in/your-profile
Create an interactive FAQ (Frequently Asked Questions) section for a personal portfolio website.
Requirements:
- Display all questions first in a clean list or cards.
- Each question should be clickable.
- When a question is clicked, show/hide the answer (toggle accordion effect).
- Only the answer of the clicked question should expand.

- Smooth animation when expanding/collapsing.
- Modern and minimal UI design.
- Mobile responsive design.

Add hover in my picture social media link, project and every where. when i scroll down the upper section will be go and when scroll up even little upper section will come.

Important rule: bold which is important for impress & attraction recruiter?

Create a professional Generative AI Engineer portfolio website for Towhid Ahmed (Bangladesh), including all sections. Fix grammar, make it recruiter-ready, and design a modern responsive UI. Add any design to make more attractive and professional. If any conflict you choose the best decisions.